

Basic Principles of Medicine 2
Module: Digestion and Metabolism
Lecture 24

Saliva and Gastric Secretions

Dr Janine Paul

jpaul@sgu.edu or ask-phy@sgu.edu

Copyright

All year 1 courses materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed or published in any form, printed, electronic or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

View only files, such as lecture recordings, are explicitly excluded from download and creating copies of these recordings by students and other users are strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St George's University. Users of this document identifying potential violations of these regulations are asked to bring their concern to the attention of the author.

Recommended Reading

Rhoades, R. & Bell, D. (2023); *Medical Physiology: Principles for Clinical Medicine*, 6th Edition; Wolters Kluwer. Chapter 25

[Gastrointestinal Secretory Functions | Medical Physiology: Principles for Clinical Medicine, 6e | Premium Basic Sciences | Health Library](#)

Objectives I

SOM.MK.II.BPM2.3.DM.1.PHYS.1117. Describe the volume and composition of salivary fluid coming from major salivary glands.

SOM.MK.II.BPM2.3.DM.1.PHYS.1118. Illustrate how acinar secretions are modified by ductal cells and predict the tonicity of the final secreted fluid depending on salivary flow rate.

SOM.MK.II.BPM2.3.DM.1.PHYS.1119. Describe the physiological function of the components of saliva.

SOM.MK.II.BPM2.3.DM.1.PHYS.1120. List the stimuli and neural pathways involved in promoting salivary secretion.

Objectives II

SOM.MK.II.BPM2.3.DM.1.PHYS.1122. Analyze the 2 types gastric luminal fluid (oxyntic and non-oxyntic) and the mechanisms that mediate between food intake and oxyntic secretion.

SOM.MK.II.BPM2.3.DM.1.PHYS.1123. Identify the proteins secreted into the gastric lumen by chief cells, parietal cells, And mucous cells. Contrast the functions and regulation of these secretions.

SOM.MK.II.BPM2.3.DM.1.PHYS.1124. Discuss the gastric cell types secreting gastrin, somatostatin, histamine, and gastrin releasing peptide.

SOM.MK.II.BPM2.3.DM.1.PHYS.1145. Discuss the stimuli that promote and inhibit release of these peptides, and their cellular targets.

SOM.MK.II.BPM2.3.DM.1.PHYS.1125. Describe the luminal pH of the stomach in the basal fasted state versus the time course of changes in luminal pH after a mixed meal.

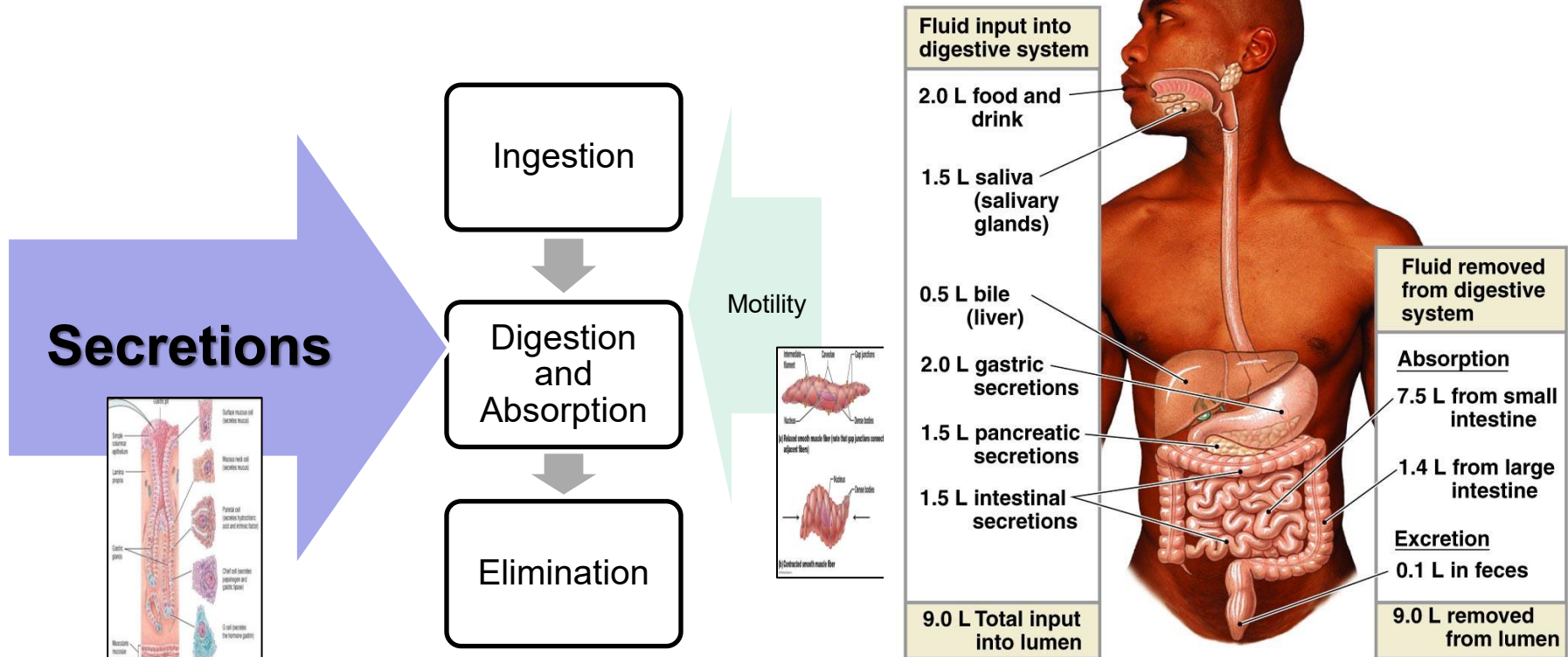
SOM.MK.II.BPM2.3.DM.1.PHYS.1126. Describe how parietal cells $H^+-K^+-ATPase$ activity can be inhibited physiologically and pharmacologically.

SOM.MK.II.BPM2.3.DM.1.PHYS.1127. Describe the ion transport mechanisms and cellular enzymes needed to allow parietal cell homeostasis during gastric acid secretion.

Outline

- Gastrointestinal secretions overview
- Secretions in the upper GI tract (mouth and stomach)
 - types of secretions
 - function
 - composition
 - regulation

GI Secretions Overview



Copyright © 2007 Pearson Education, Inc., publishing as Benjamin Cummings.

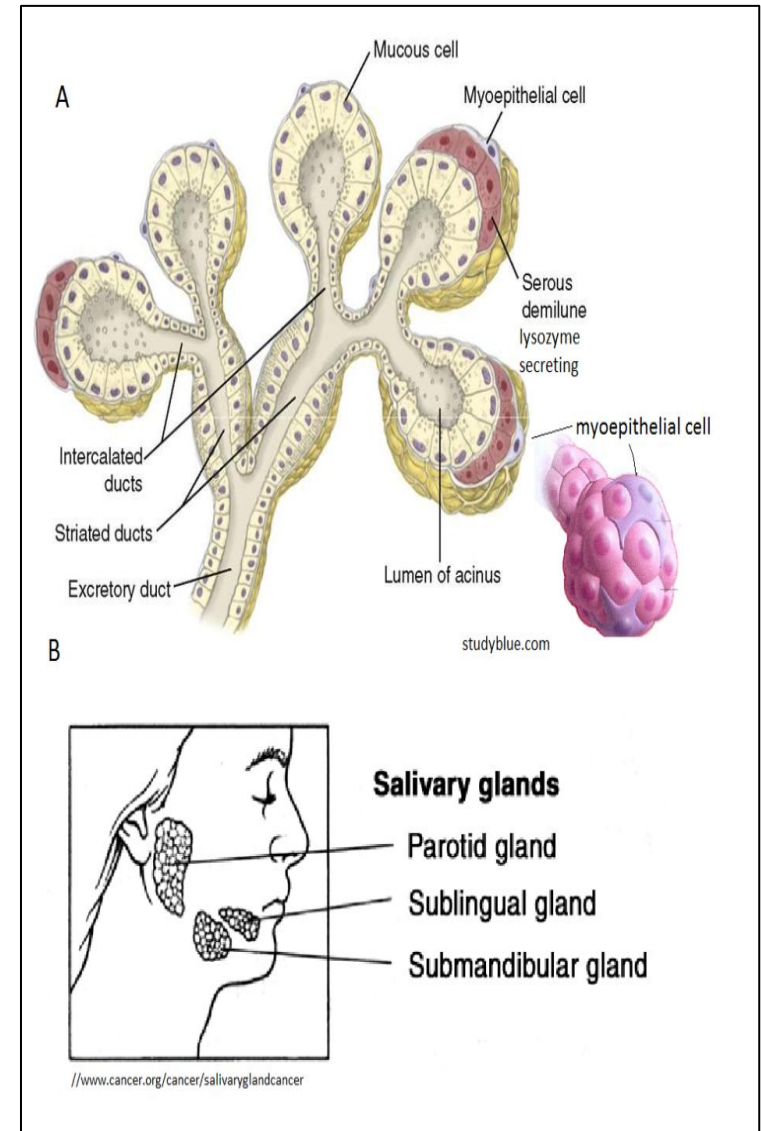
- Secretions: release of digestive juices into the lumen of the GIT
- Secretions aid in digestion and absorption
- Regulation is both neural and hormonal
- ~9 L pass through GIT daily: ~2 L ingested + ~7 L secretions
- 99% of luminal content is reabsorbed

Part 1

Oral cavity secretions

Oral cavity - Saliva

- Composition:
 - From Parotid, Submandibular and Sublingual glands
 - 99% water (large volume)
 - 1% salts, enzymes (amylase, lipase) and mucins
 - Hypotonic
- Function :
 - Protection
 - Taste sensation
 - Lubrication and helps articulation
 - Digestion
- Control of salivation is mediated almost entirely by nervous system
- Both Sympathetic and Parasympathetic stimulate secretion



Acinar secretion & ductal modification

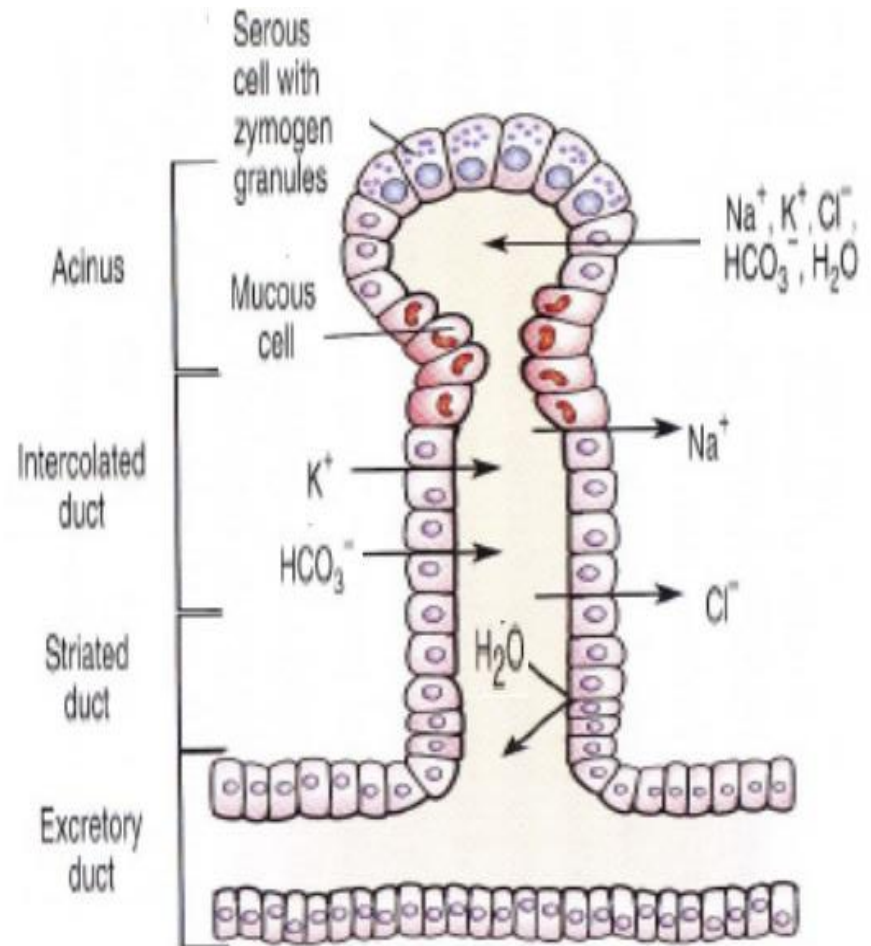
Step 1

The acinus produces an initial saliva similar to plasma (isotonic). Contains Na^+ , K^+ , Cl^- , HCO_3^-

Step 2

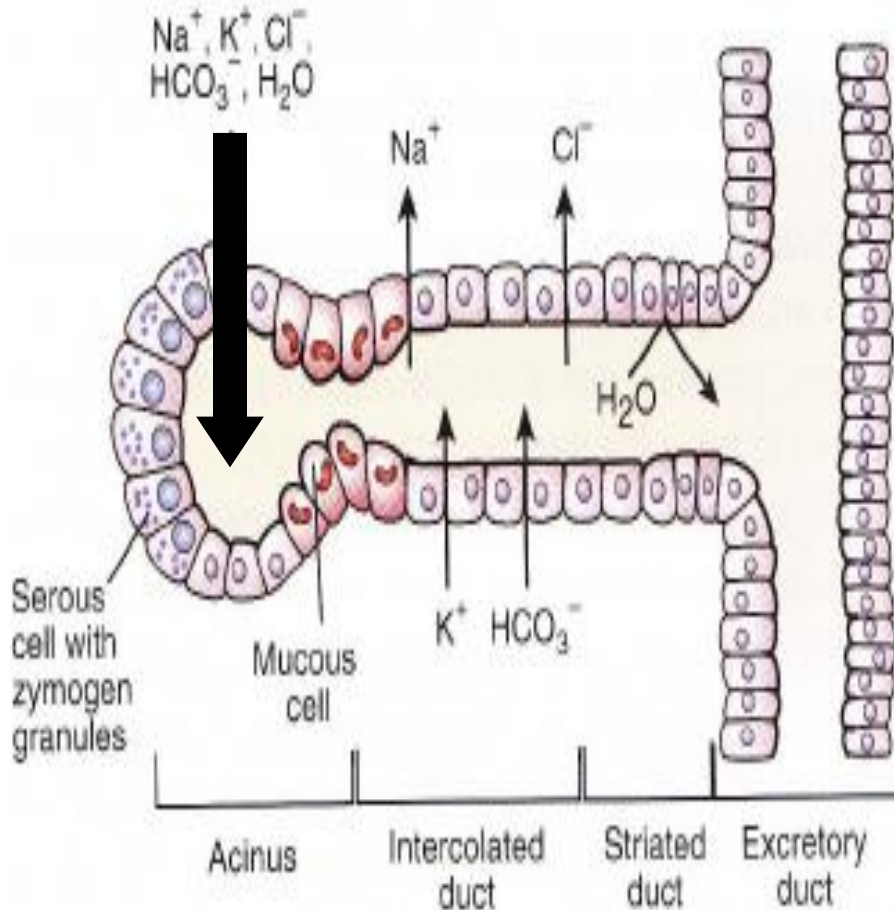
Ducts modify the saliva: absorb Na^+ , Cl^- and secrete K^+ and HCO_3^- (hypotonic)

At low flow rates (basal), contact time allows for reabsorption and secretion. At high flow rates (stimulated) saliva becomes more similar to acinar secretions.



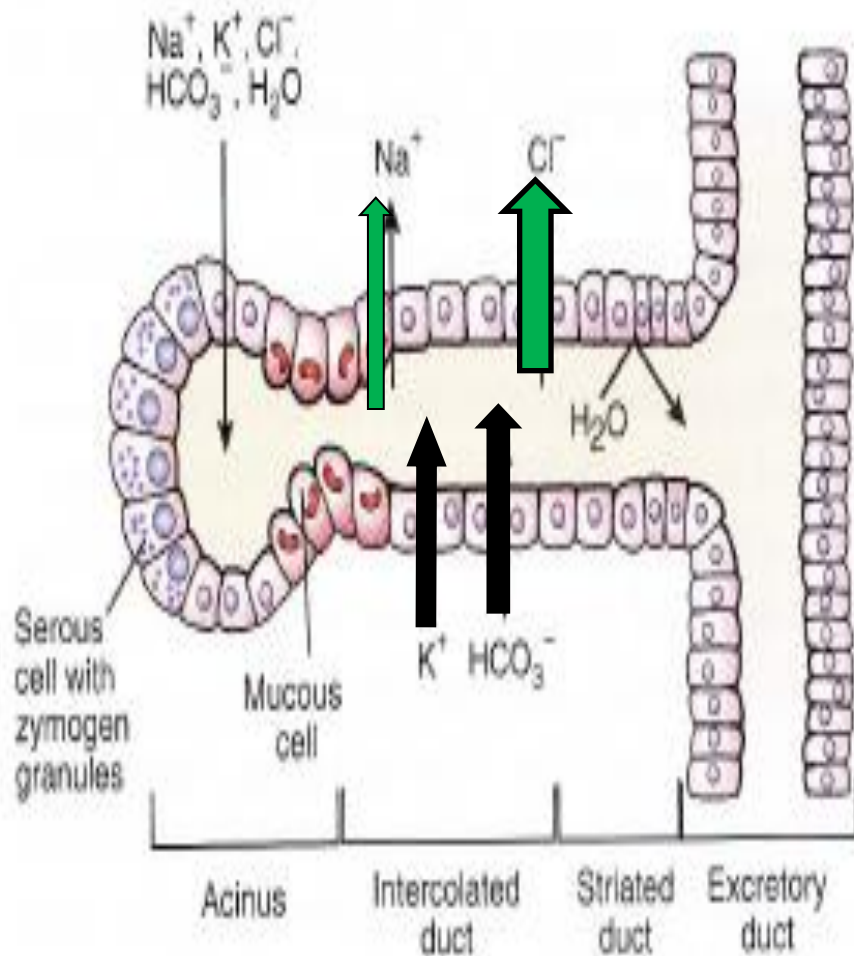
Acinar secretion

The Salivon



- acinar cells are **leaky** and secrete digestive enzymes, mucin, electrolytes and water (isotonic NaCl)
- Cl is co-transported with Na into the cell basolaterally
- Cl diffuses down the gradient into the lumen (apical)
- channel allows HCO_3^- to enter the lumen
- Na^+ and H_2O follow paracellularly
- extent of modification depends on flow rate

Ductal modification

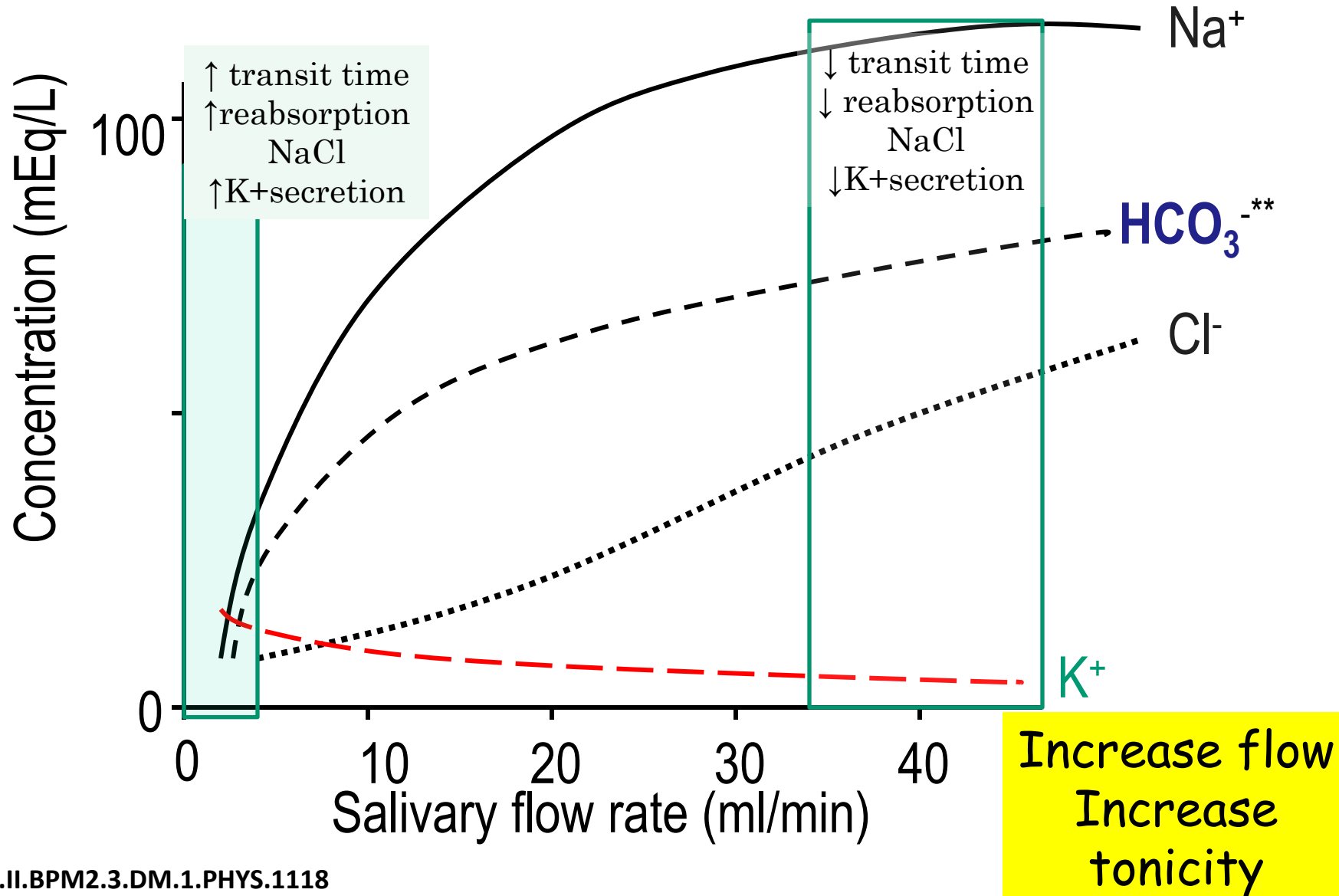


2.

HYPOTONIC

- Ductal cells are **tight** (low water permeability)
- Acinar secretion is modified in the duct
 - K^+ and HCO_3^- are actively secreted
 - Na-K ATPase on BL membrane drives active reabsorption of Na^+ and K^+ secretion
 - anionic exchanger reabsorbs Cl^-
 - active secretion HCO_3^- via anionic exchanger ($\text{Cl}^-/\text{HCO}_3^-$)

Flow rate affects saliva composition



Summary of Saliva production

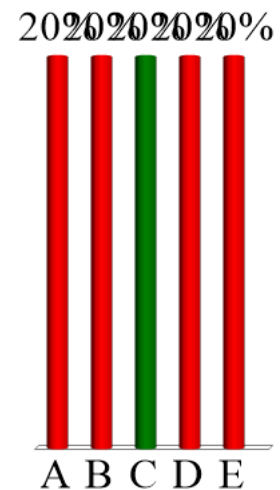
A primary, isotonic secretion is formed in the acinum. Acinar cells secrete Cl , HCO_3 and K . Cl is co-transported with sodium into the cell at the basolateral membrane. Cl then diffuses down the concentration gradient into the acinus via an electrogenic channel which also allows HCO_3 flux. Acinar cells are leaky, so Na and water follow paracellularly.

The ductal cells then modify the initial acinar secretion by actively secreting K^+ and absorbing Na^+ from the lumen. This process depends on concentration gradients driven by the Na-K ATPase at the basolateral membrane. Similarly, there is an anionic exchange with Cl reabsorbed from saliva and HCO_3 secreted.

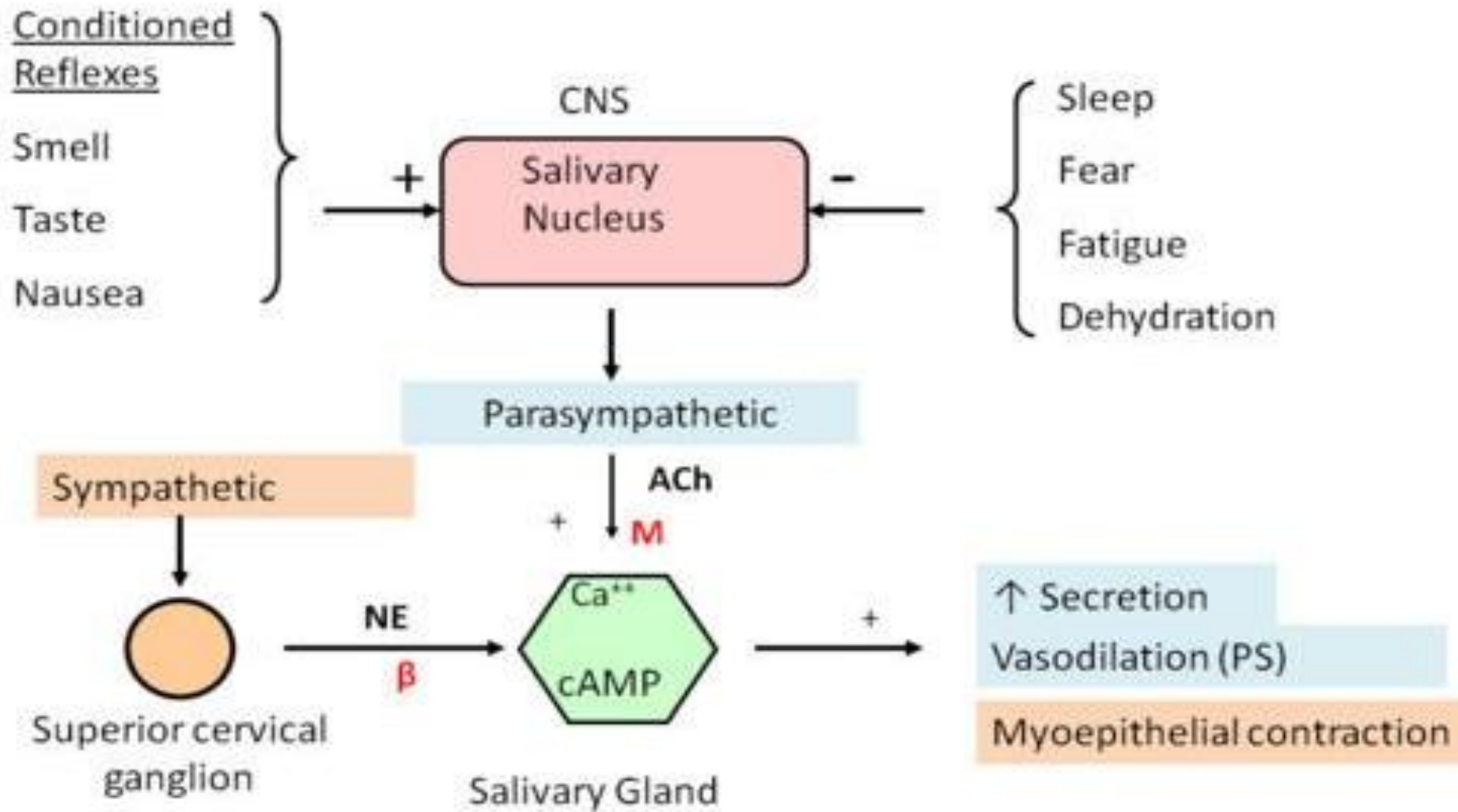
At low secretion rates, the ductal epithelium has more time to modify and reduce the osmolality of the primary secretion, whereas at high flow rates it has an osmolality approaching plasma. Saliva is hypotonic to plasma because of a net absorption of ions by the ductal epithelium. More Na and Cl is absorbed than K and HCO_3 being secreted, hence there is net absorption of solute. The epithelial lining of the duct has a low permeability to water, so water does not follow the absorbed salt.

Investigators conduct a study that evaluates the effect of rate of flow on the composition of the saliva, using 50 healthy volunteers. At rest, with a low salivary flow rate, a hypotonic saliva is collected that contains high concentrations of potassium and bicarbonate and low concentrations of sodium and chloride. Salivary glands are stimulated with a cholinergic agonist to increase the secretion and the flow rate. Which of the following alterations in the composition of saliva is most likely after stimulation?

	Sodium (Na ⁺)	Potassium (K ⁺)	Bicarbonate	Chloride (Cl ⁻)
A.	Decreases	Increases	Increases	Decreases
B.	Increases	Decreases	Decreases	Increases
✓ C.	Increases	Decreases	Increases	Increases
D.	Decreases	Decreases	Increases	Decreases
E.	Increases	Increases	Decreases	Decreases



Regulation of saliva production



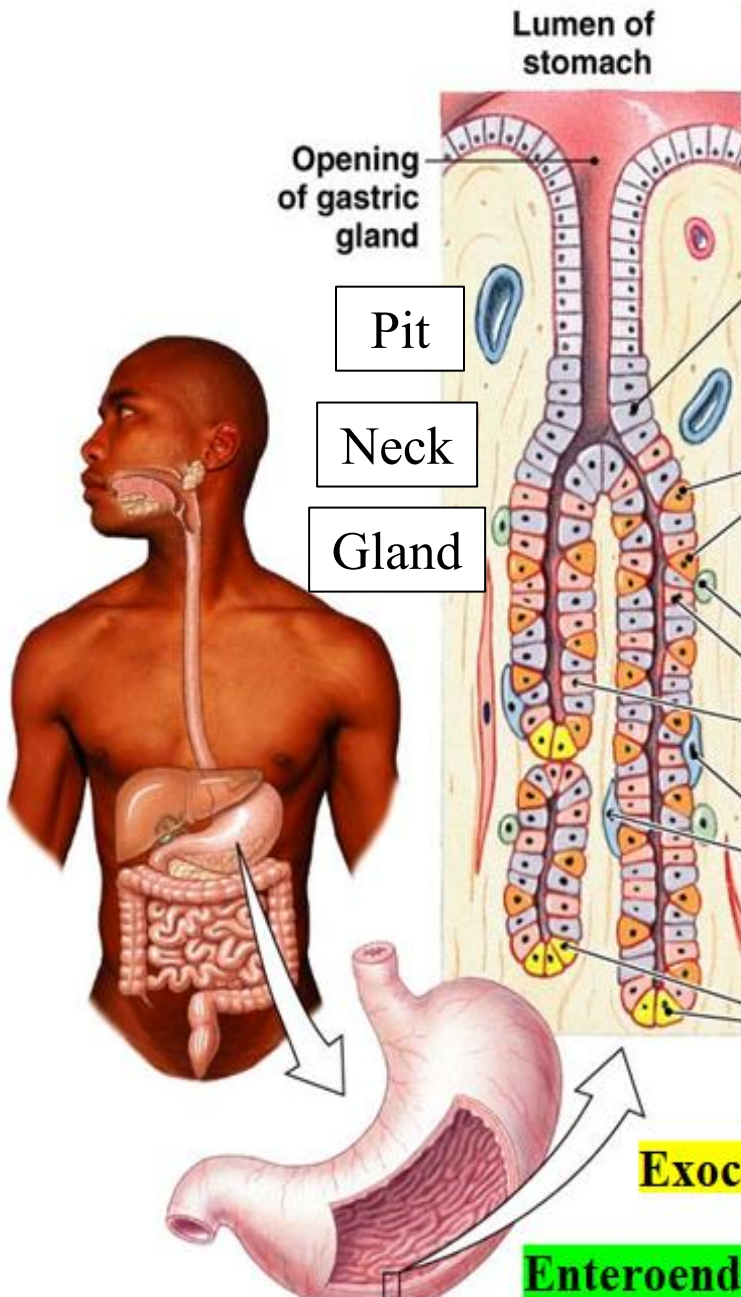
Regulation of Salivary Secretion

- Both parasympathetic (glossopharyngeal - parotid, facial nerve - submandibular and sublingual glands) and sympathetic ANS regulate release of saliva
- Parasympathetic more **predominant**
- Mechanisms of control
 - Parasympathetic stimulation produces a secretion rich in electrolytes and salivary amylase (enzyme).
 - Sympathetic stimulation produces a secretion rich in mucus.
 - Ach, NE, VIP and substance P are the neurotransmitters involved.
- Resting state salivary secretion
 - low at 30 ml /hr
 - submandibular glands contributing ~ 2/3 resting saliva
- Stimulated glands
 - secrete up to 400 ml /hr
 - mostly from parotid gland

Part 2

Gastric secretions

Summary: Gastric secretions



Cell Types	Substance Secreted	Stimulus for Release	Function of Secretion
Mucous neck cell	Mucus	Tonic secretion; with irritation of mucosa	Physical barrier between lumen and epithelium
	Bicarbonate	Secreted with mucus	Buffers gastric acid to prevent damage to epithelium
Parietal cells	Gastric acid (HCl)	Acetylcholine, gastrin, histamine	Activates pepsin; kills bacteria
	Intrinsic factor		Complexes with vitamin B ₁₂ to permit absorption
Enterochromaffin-like cell	Histamine	Acetylcholine, gastrin	Stimulates gastric acid secretion
Chief cells	Pepsin(ogen)	Acetylcholine acid, secretin	Digests proteins
	Gastric lipase		Digests fats
D cells	Somatostatin	Acid in the stomach	Inhibits gastric acid secretion
G cells	Gastrin	Acetylcholine, peptides, and amino acids	Stimulates gastric acid secretion

Exocrine secretions

Enteroendocrine secretions

Pernicious anemia

Definition:

Pernicious anemia is a type of **megaloblastic anemia** caused by autoimmune destruction of gastric parietal cells, leading to vitamin B12 deficiency.

Pathophysiology:

1. Gastric Parietal Cell Damage:

- Autoimmune destruction of parietal cells (located in the stomach's fundus/body) leads to:
 - ❖ Decreased **intrinsic factor** (IF), which is essential for vitamin B12 absorption.
 - ❖ Decreased gastric acid secretion (**achlorhydria**).

2. Vitamin B12 Absorption Failure:

- Vitamin B12 binds to intrinsic factor in the stomach.
- This complex is absorbed in the **terminal ileum** via IF receptors.
- Lack of intrinsic factor results in vitamin B12 malabsorption.

3. Impaired DNA Synthesis:

- Vitamin B12 is essential for DNA synthesis (via folate metabolism).
- Deficiency causes impaired RBC maturation, leading to **megaloblastic anemia**.

Gastric Secretions – 1-2L/day

- Composition changes with flow rates
 - Cellular mechanism H^+ secretion
 - Factors that increase and decrease secretion
 - Amounts secreted vary during each GI Event:
cephalic, gastric, intestinal phase

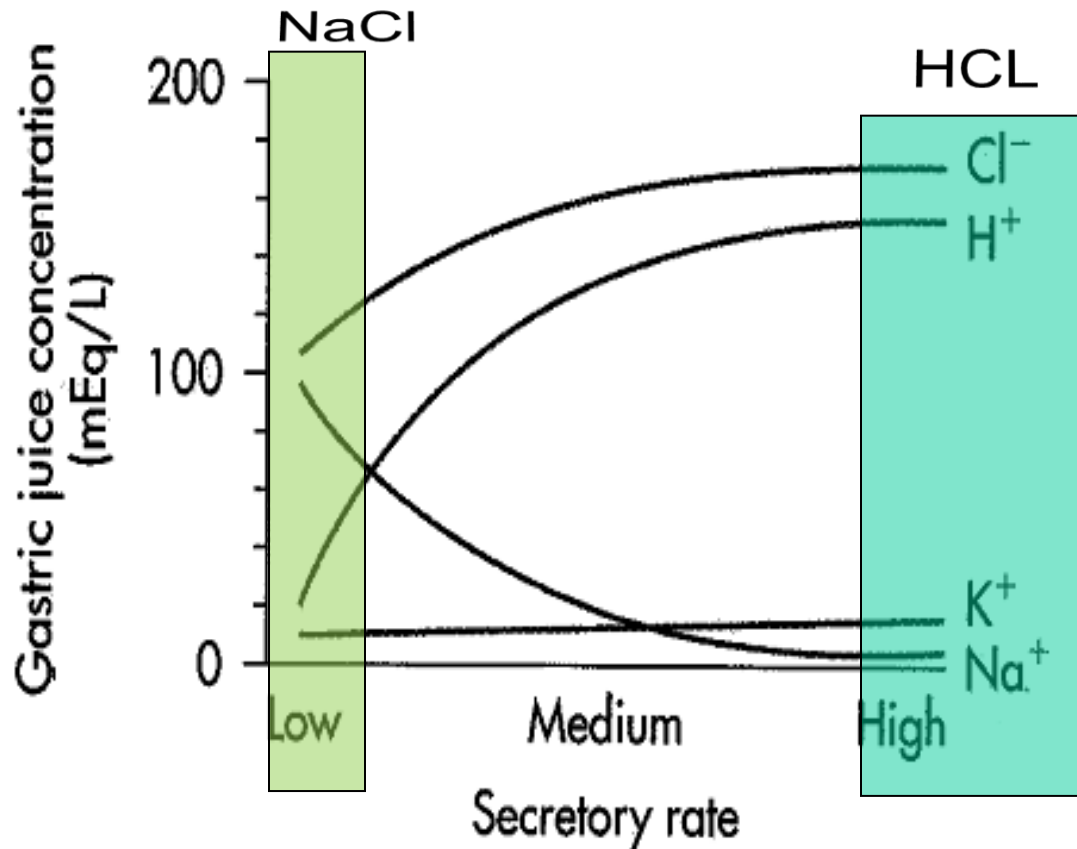
Basal Secretion : similar to ECF

- Mainly mucus and water (paracellular diffusion of fluid from ECF and secretions from mucus glands)
- Non-oxyntic component
- Isotonic (high Na^+ and Cl^- content)

Stimulated Secretion : change

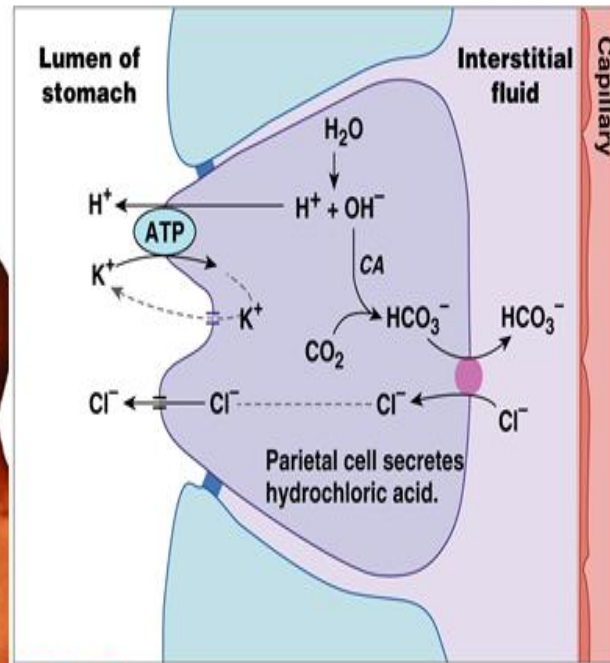
- Parietal cells secrete HCL
- Acidic
- Oxyntic component
- Still Isotonic (high H^+ and Cl^- content)

Based on the two-component model,
what variations do you expect?



	Oxyntic fluid	Non-oxyntic fluid
Na^+	5	140
K^+	20	5
H^+	140	10^{-4}
Cl^-	165	110
HCO_3^-	0	25

Mechanism of Gastric Acid Secretion



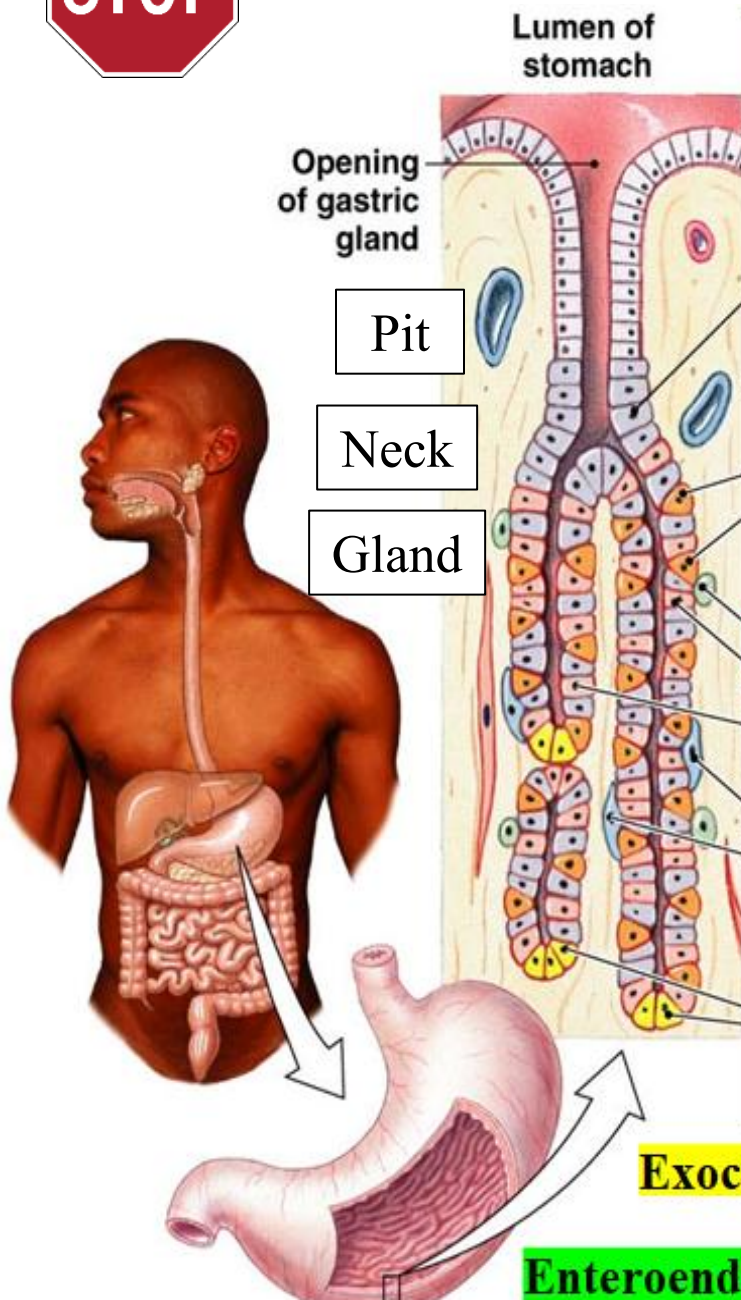
1. $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}^+ + \text{HCO}_3^-$
(catalyst carbonic anhydrase)
2. H^+ is secreted into the lumen via H^+/K^+ ATPase
3. K^+ can leak back into lumen
4. Cl^- is also secreted
5. HCO_3^- produced absorbed into the blood stream in exchange for Cl^- ($\text{Cl}^-/\text{HCO}_3^-$ exchanger)
6. **alkaline tide:** increase in venous blood pH
7. Tight junctions prevent paracellular transport
8. 3 stimulants for parietal cells:
ACh, gastrin and histamine

Gastric acid summary

- Basal secretion is from non-parietal cells, high Na^+ and Cl^- (resembles plasma)
- Stimulation of parietal cells → production of acid secretion
- Composition now becomes high in H^+ , K^+ , and Cl^- is also elevated
- Gastric juice becomes acidic
- Overall effect of acid secretion is H^+ into stomach lumen, and HCO_3^- into venous blood. pH of venous blood rises: alkaline tide



Summary: Gastric secretions



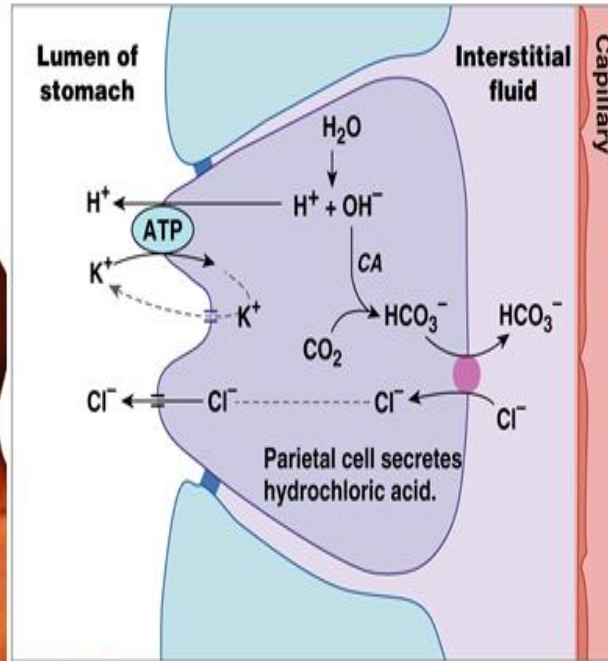
Cell Types	Substance Secreted	Stimulus for Release	Function of Secretion
Mucous neck cell	Mucus	Tonic secretion; with irritation of mucosa	Physical barrier between lumen and epithelium
	Bicarbonate	Secreted with mucus	Buffers gastric acid to prevent damage to epithelium
Parietal cells	Gastric acid (HCl)	Acetylcholine, gastrin, histamine	Activates pepsin; kills bacteria
	Intrinsic factor		Complexes with vitamin B ₁₂ to permit absorption
Enterochromaffin-like cell	Histamine	Acetylcholine, gastrin	Stimulates gastric acid secretion
Chief cells	Pepsin(ogen)	Acetylcholine acid, secretin	Digests proteins
	Gastric lipase		Digests fats
D cells	Somatostatin	Acid in the stomach	Inhibits gastric acid secretion
G cells	Gastrin	Acetylcholine, peptides, and amino acids	Stimulates gastric acid secretion

Exocrine secretions

Enteroendocrine secretions



Mechanism of Gastric Acid Secretion

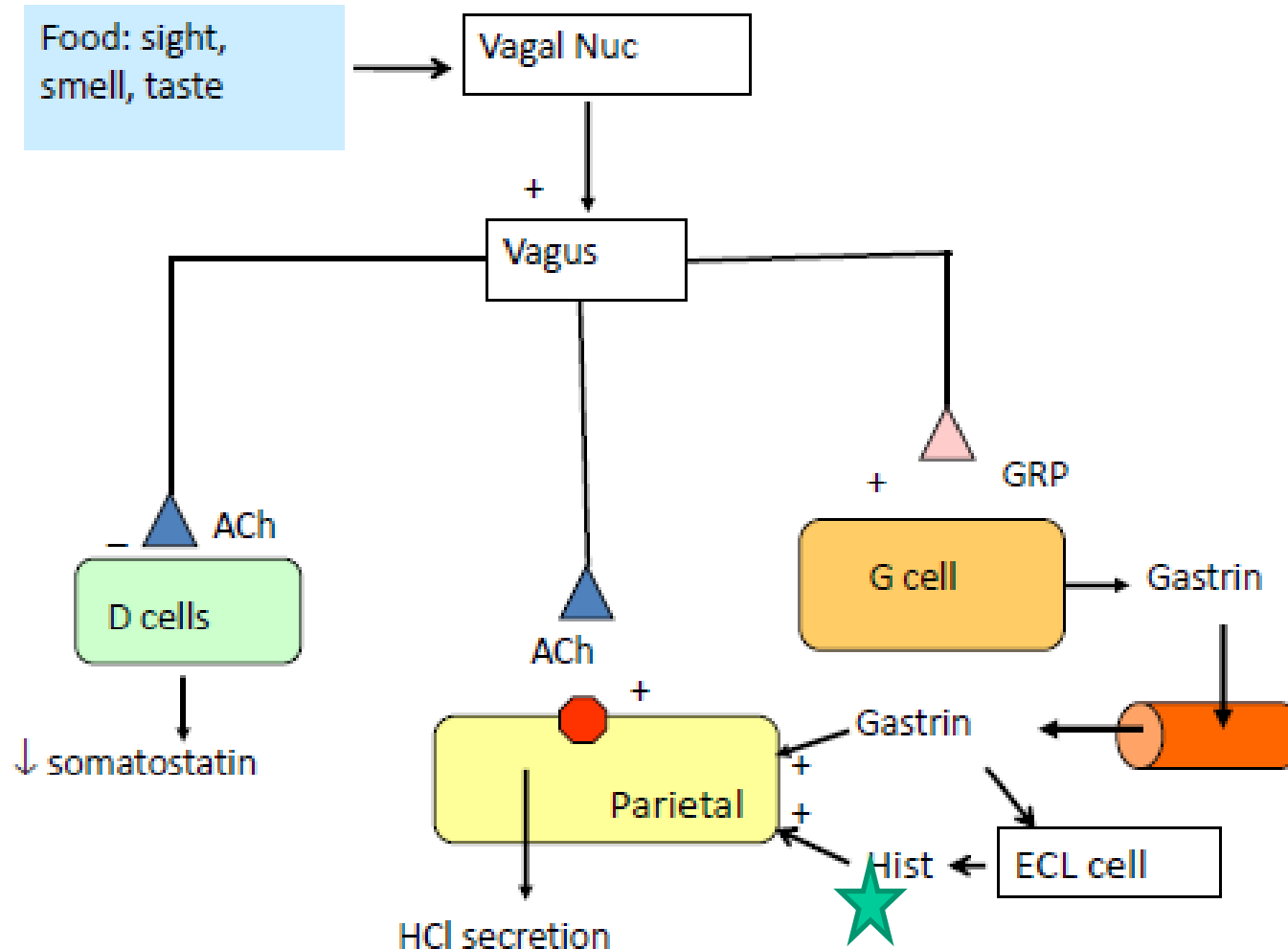


1. $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}^+ + \text{HCO}_3^-$
(catalyst carbonic anhydrase)
2. H^+ is secreted into the lumen via H-K ATPase
3. K^+ can leak back into lumen
4. Cl^- is also secreted
5. HCO_3^- produced absorbed into the blood stream in exchange for Cl^- (Cl^- - HCO_3^- exchanger)
6. **alkaline tide:** increase in venous blood pH
7. Tight junctions prevent paracellular transport
8. 3 stimulants for parietal cells:
 ACh , gastrin and histamine

Timing of Gastric Secretions

Phase	Stimulus	Pathway	Stimulus to Parietal Cell
Cephalic 30%	Thought of food, smell, taste, chewing and swallowing	Vagus nerve to Parietal cells and G cells	Ach and Gastrin (from G cells)
Gastric 50%	Stomach distension by food	Local ENS reflexes and vagovagal reflexes to Parietal cells and G cells	Ach and Gastrin
Intestinal 5%	Protein digestion products in duodenum Distension	Amino acids in blood. Intestinal endocrine cell	Amino acids and Entero-oxyntin

Cephalic phase acid production

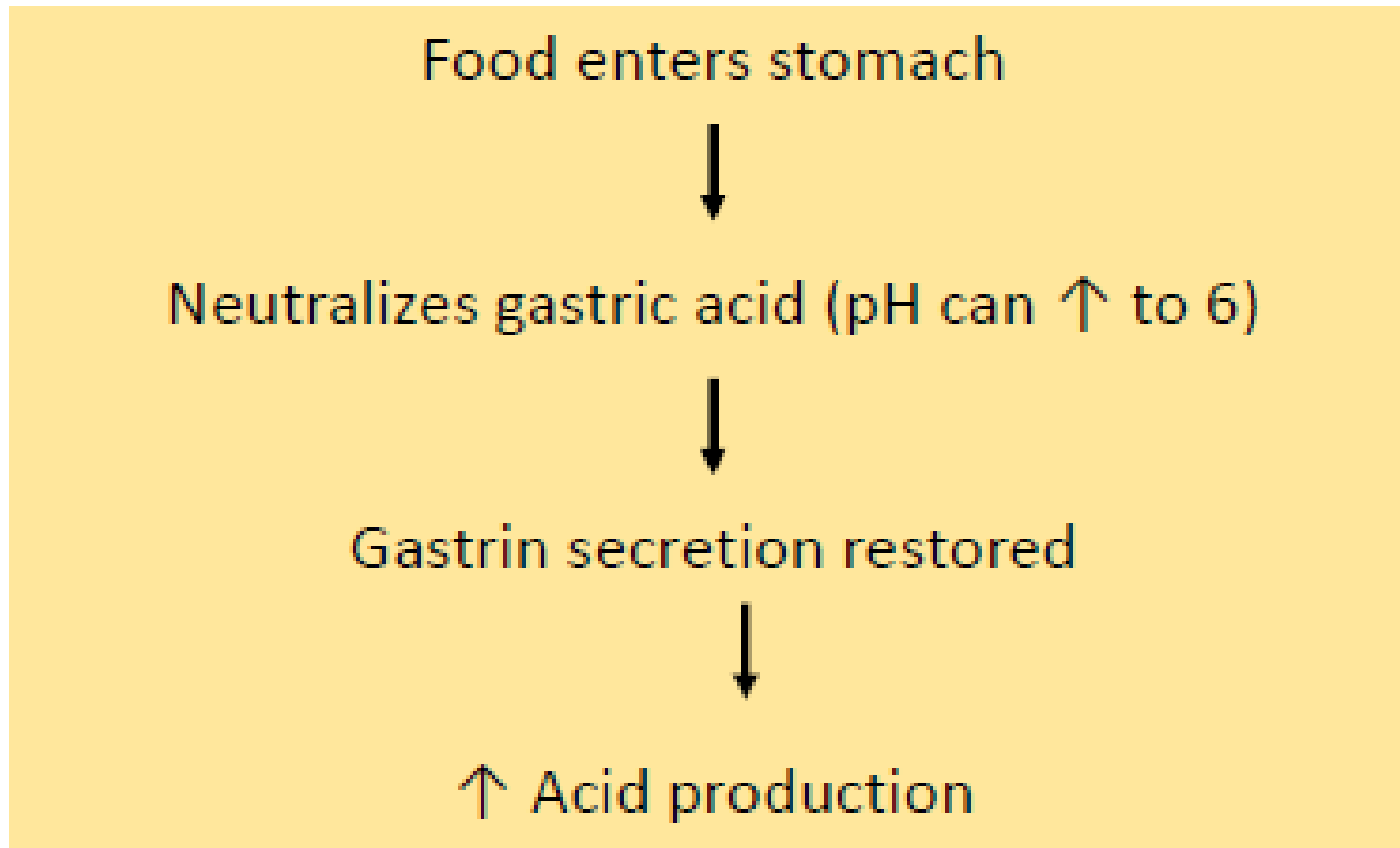


Necessary to inhibit somatostatin since SST normally ↓ gastrin

Cephalic phase acid production

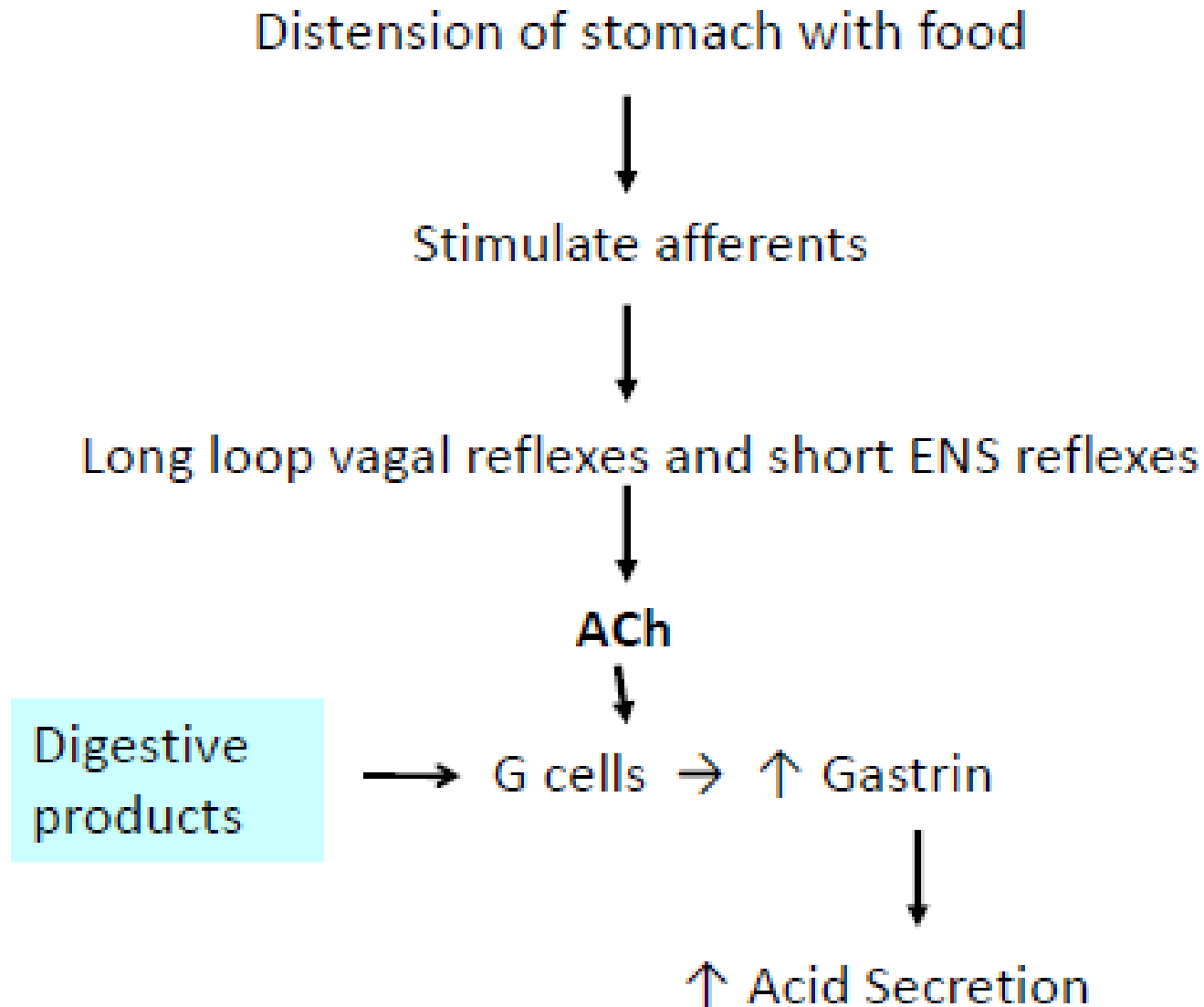
The most important stimulus for gastric acid secretion is direct release of Ach by nerve terminals on oxyntic cells. The second mechanism involves release of gastrin. Post-ganglionic vagal efferents ending on G cells release Gastrin Releasing Peptide (GRP) as a neurotransmitter, causing gastrin secretion into the blood. At the same time cholinergic vagal efferents end on antral D cells inhibit somatostatin secretion, which is necessary because somatostatin normally inhibits the release of gastrin via a paracrine mechanism.

Gastric phase acid production



50- 60% of gastric acid is secreted during gastric phase

Stomach stretch & acid production



Stimulation Of Acid Secretion

Interdigestive phase	15%
Cephalic phase	30%
Gastric phase	50%
Intestinal phase	5%

POTENTIATION!

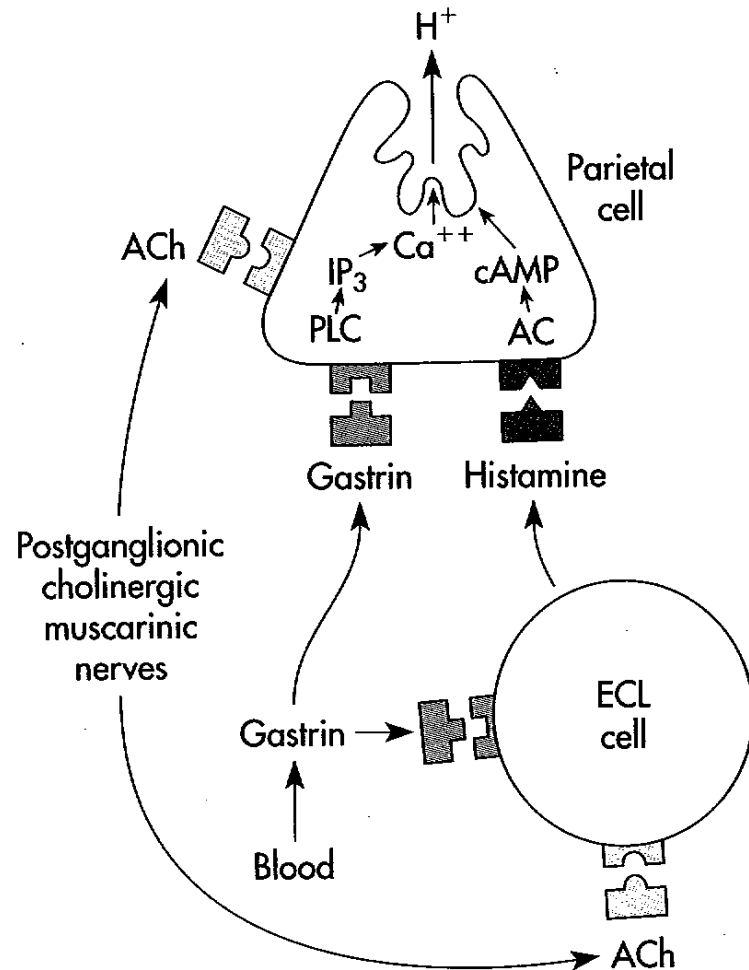
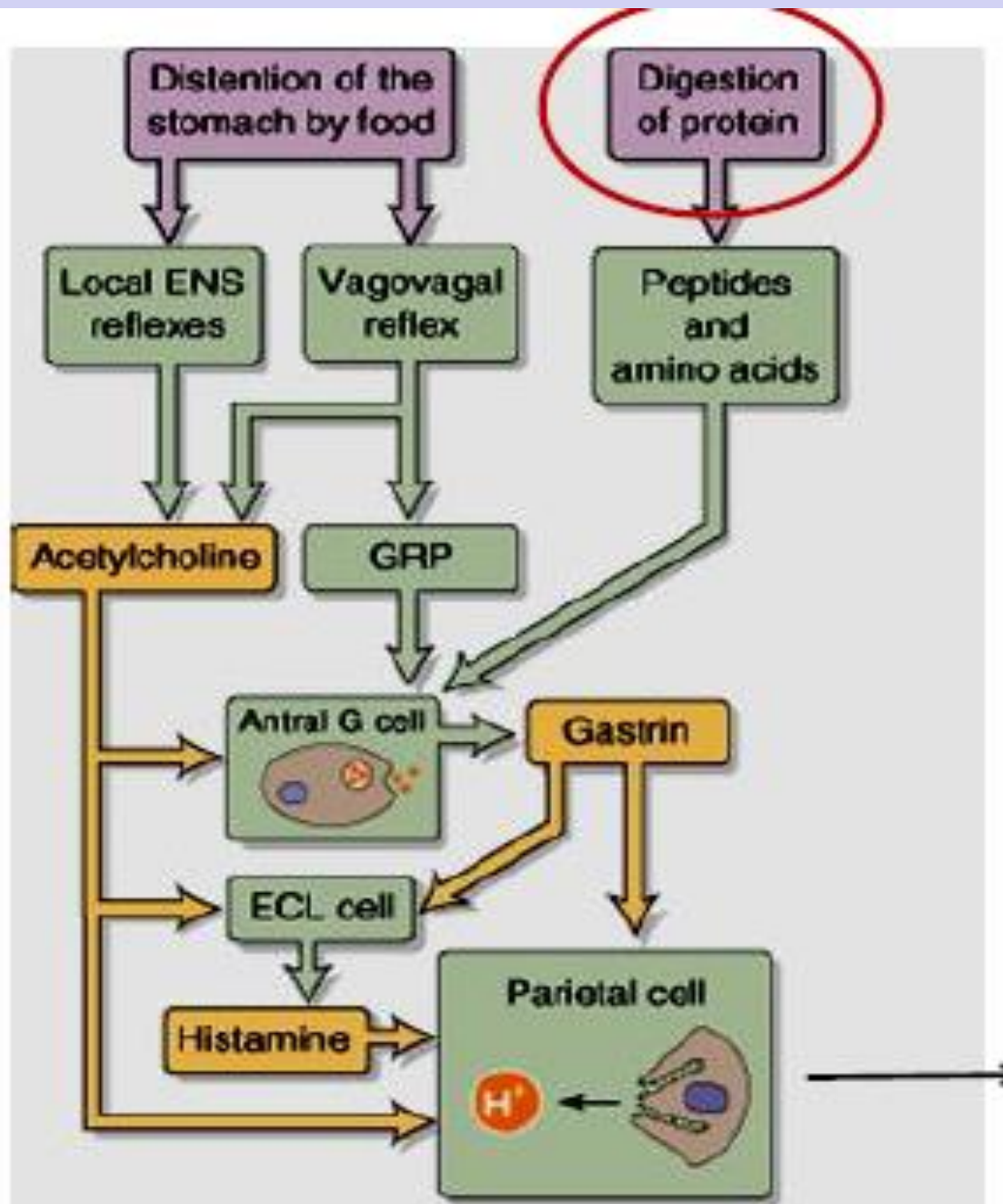


Diagram after L Johnson

Overview: Regulation of gastric acid secretion



Summary: Phases of gastric secretion

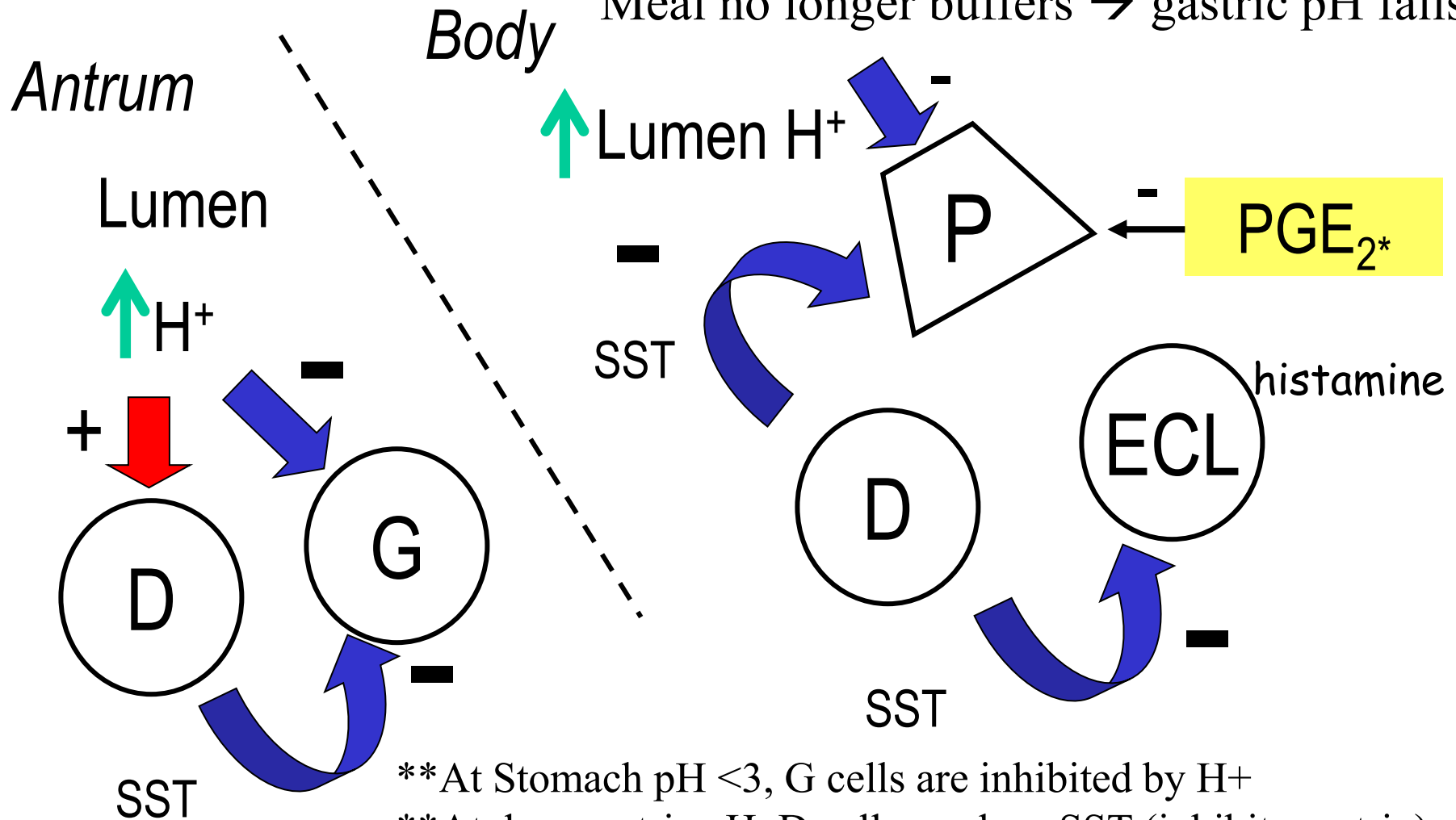
Cephalic phase: The cephalic phase of secretion normally accounts for ~ **30% of gastric secretion**, before the entry of any food into the stomach. The neurogenic stimuli for this phase originate in the cerebral cortex and appetite centres and reach the stomach via the vagus nerves. The vagal nerves release ACh, which stimulates parietal cell H^+ secretion. ACh also causes histamine release from Enterochromaffin like (ECL) cells. The vagus nerve also releases gastrin releasing peptide (GRP), which causes gastrin to be released from antral G cells. Cholinergic vagal efferents inhibit antral D cells and therefore reduce the production of somatostatin, which would otherwise inhibit gastrin release.

Gastric Phase: The gastric phase is responsible for **50 to 60% of the total gastric secretion**. Food entering the stomach causes gastric acid secretion by two major mechanisms: mechanical stretch and the presence of partially digested proteins. Distension of the stomach activates a vagovagal reflex and the local enteric nervous system (ENS) pathway. The partially digested proteins stimulate antral G cells to release gastrin.

Intestinal Phase: **5-10% of the total gastric secretion** comes from the intestinal phase. Partially digested peptides and amino acids in the proximal part of the small intestine activate the duodenal G cells to produce gastrin. Distension of small intestine, probably via the release of the hormone entero-oxyntin from intestinal endocrine cells, stimulates acid secretion

Negative Feedback of Acid Secretion

Most gastric acid release ~1hr after meal.
Meal no longer buffers $\rightarrow$ gastric pH falls



**At Stomach pH <3, G cells are inhibited by H^+

**At low gastric pH, D cells produce SST (inhibits gastrin)

Inhibition of gastric acid secretion

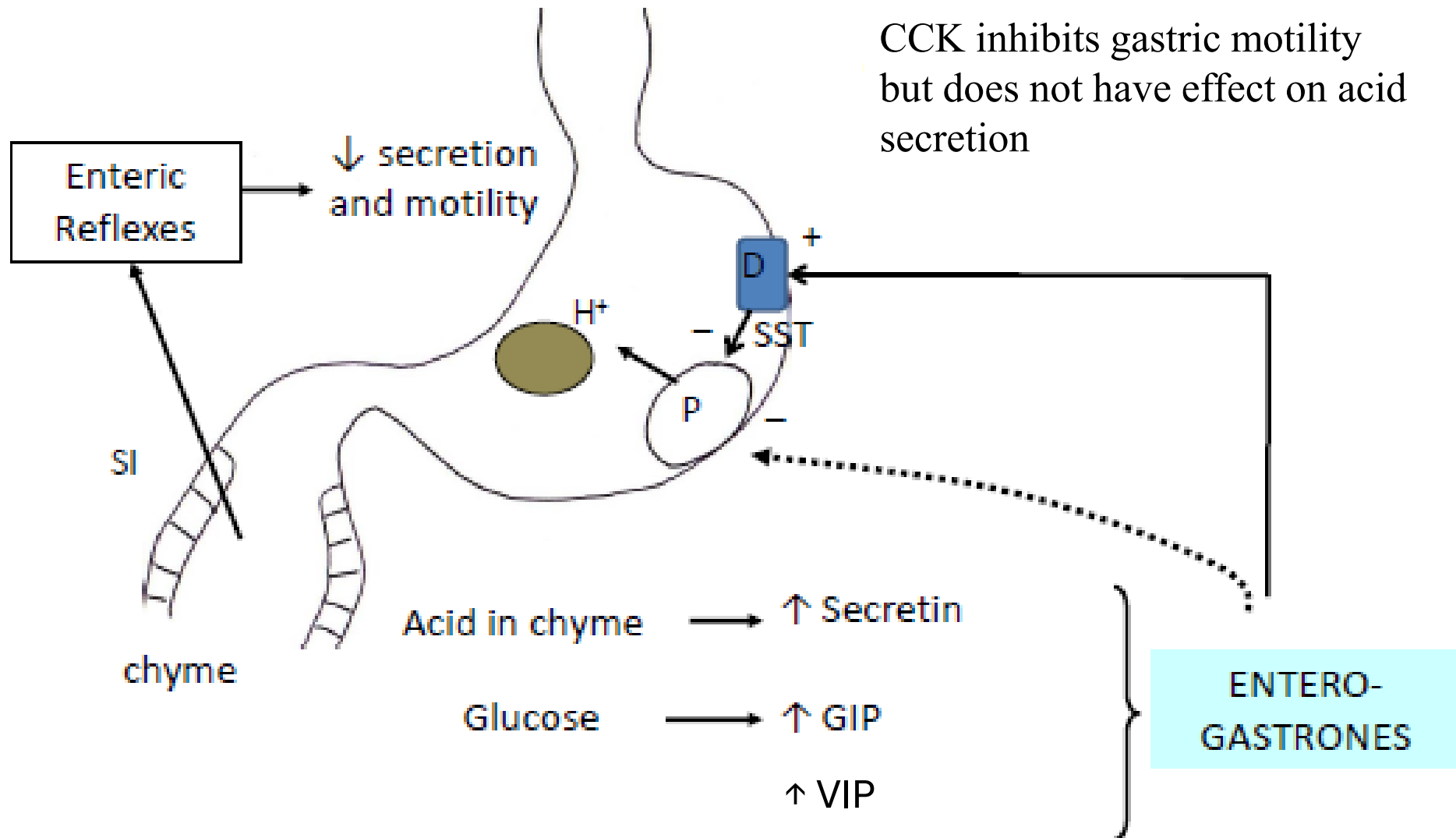
The inhibition of gastric acid secretion is important for two reasons. 1. Secretion of acid is only important during the digestion of food. 2. Excess acid causes mucosal damage. After a meal the proteins in the food buffer the acidity and the pH is usually >3 . If the protein buffering capacity is exceeded, or the stomach is empty, pH falls <3 and the G cell is directly inhibited by H^+ . Falling pH also stimulates D cells in the antrum to release somatostatin, which inhibits gastrin. Low pH also inhibits oxyntic cells.

Stomach pH is the most sensitive regulator of acid secretion. Somatostatin released by gastric D cells in the antrum is the central mechanism of inhibition of acid secretion. Somatostatin inhibits gastric acid secretion by both direct and indirect pathways. Direct pathway: somatostatin binds directly to the parietal cell. The source of this SST can be paracrine, D cells in the corpus or endocrine D cells in the antrum. Indirect pathways (both paracrine): In the corpus D cells release SST, which inhibits release of histamine from ECL cells. In the antrum D cells release SST, which inhibits the release of gastrin from G cells.

During the interdigestive period the stomach only secretes a few mls of almost entirely nonoxyntic fluid composed mostly of mucous, with little pepsin, and pH can range from 3 to 7. The pH of the fluid in the empty (fasted) stomach is usually <2 because of the basal H^+ secretion and the absence of food that would otherwise buffer the secreted acid.

Inhibition from the SI: Enterogastrones

Enterogastrones are enteric Hormones inhibiting gastric functions



How can drugs reduce acid secretion?

1. H-K ATPase inhibitors	e.g. Omeprazole, Lanzoprazole
2. Histamine receptor blockers (H2-blockers)	e.g. Cimetidine, Ranitidine
3. Muscarinic antagonist	Atropine (not in clinical use)
4. Gastrin receptor blocker	Netazepide (experimental)

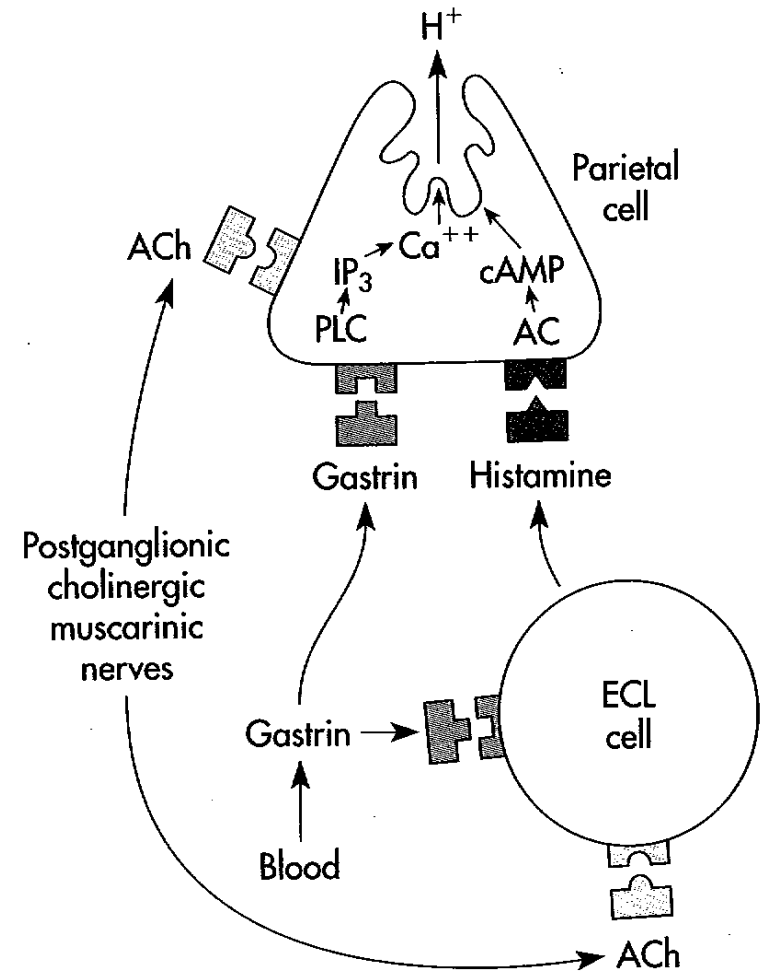
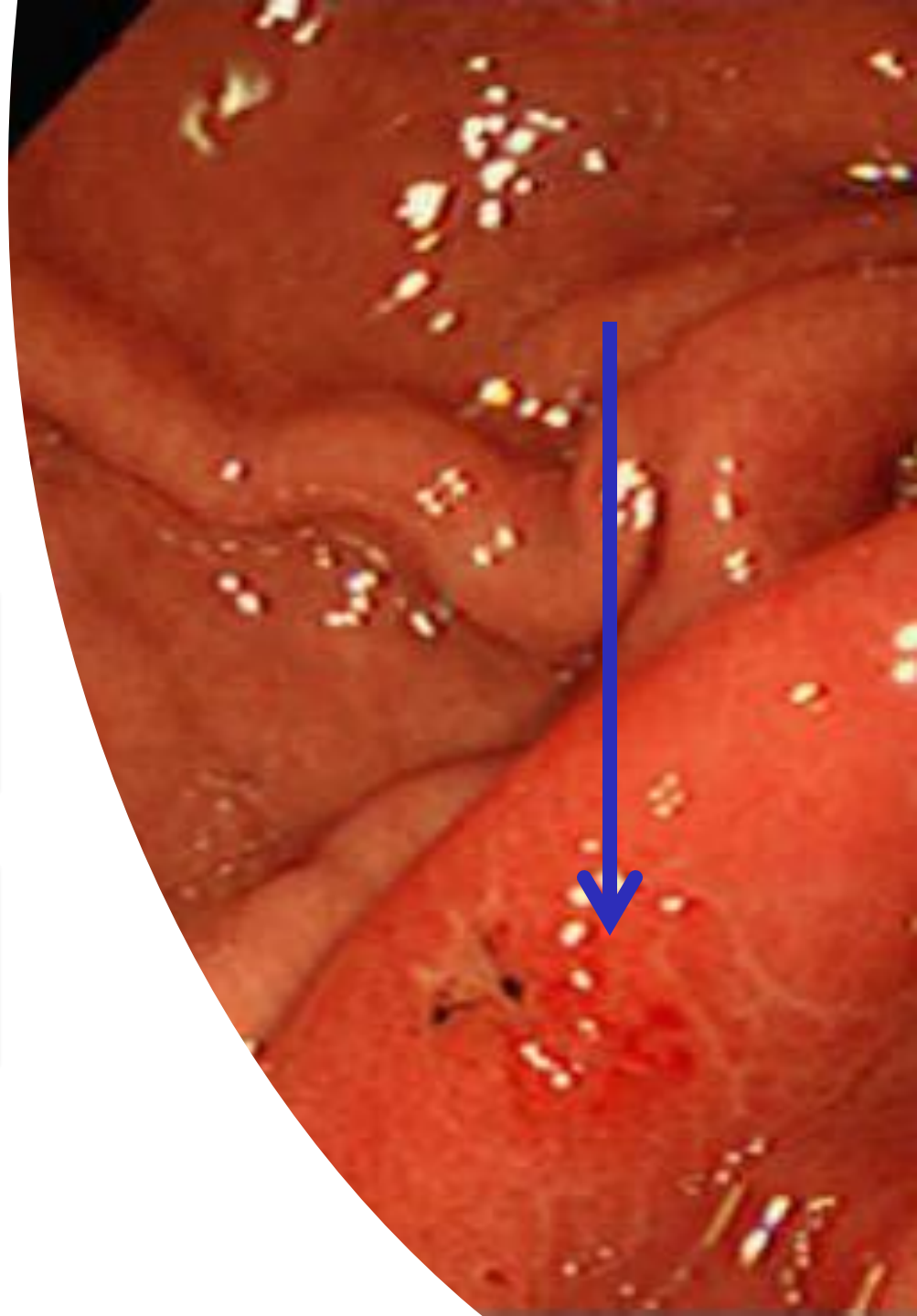
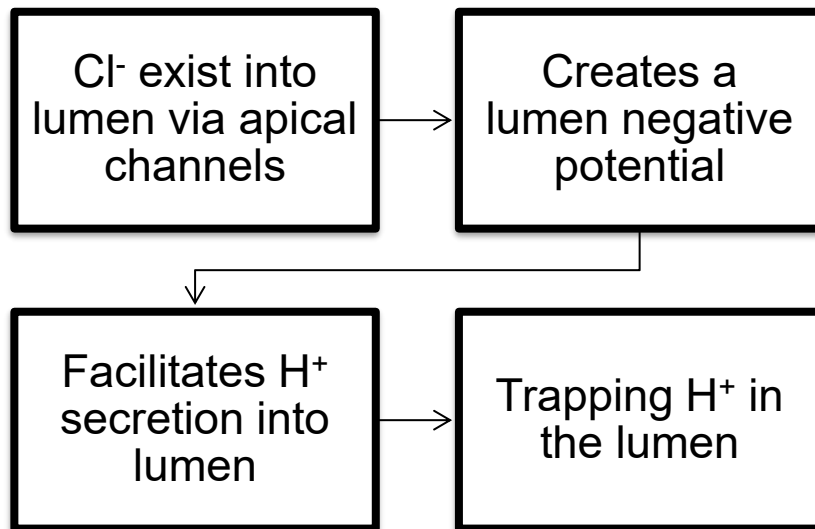
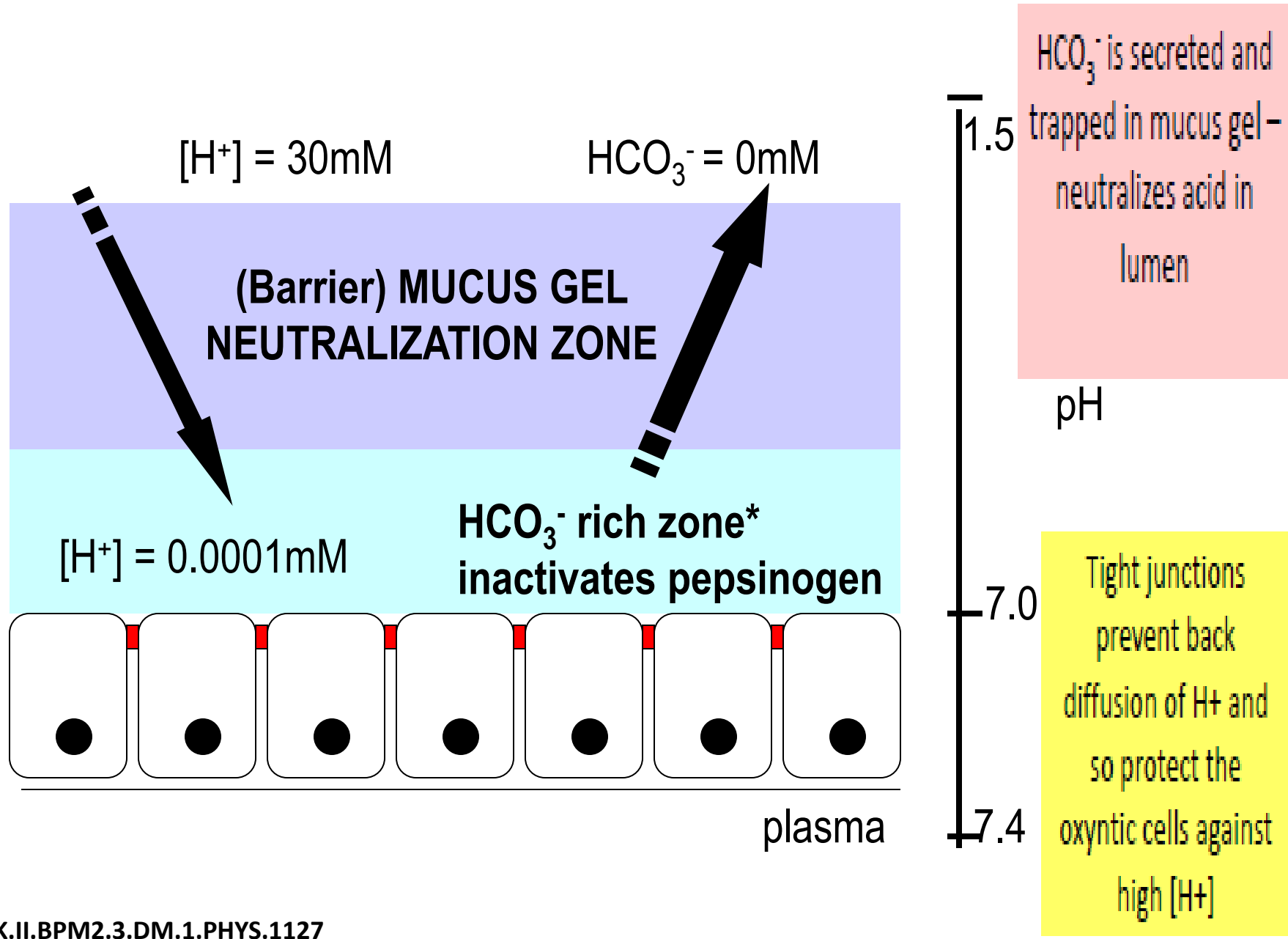


Diagram after L Johnson

The Gastric Mucosal Barrier protects against damage from gastric acid



The Gastric Mucosal Barrier



The Gastric Mucosal Barrier

Secreted mucus forms a mucous gel barrier through which H^+ cannot diffuse back through it

Between mucus layer and epithelial cell is a HCO_3^- rich fluid. Any H^+ that diffuses back from lumen is neutralized by HCO_3^- .

The HCO_3^- solution also inactivates pepsinogen, preventing pepsin from digesting cells

Prostaglandins stimulate HCO_3^- production by gastric mucosa and maintain the gastric mucosa barrier.

Gastro(duodenal) barrier integrity

HCO_3^-

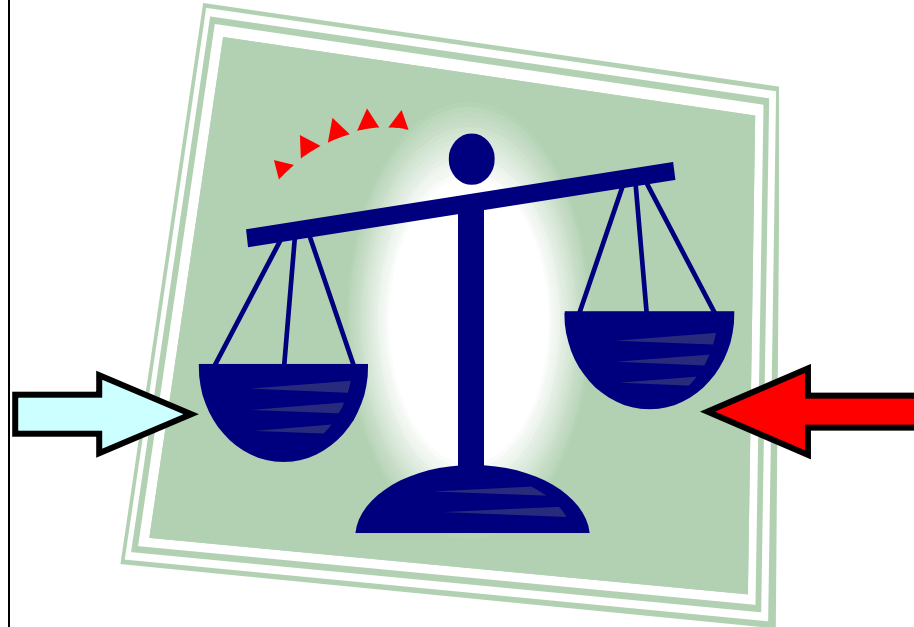
Mucus

Blood flow

Growth factors

Cell Renewal

Prostaglandins,
 HCO_3^- , $\uparrow$ blood
flow



H^+ , pepsins

Ethanol*

NSAIDs*

Bile acids

Ischemia

Smoking

H. Pylori

Ulcers form when there is imbalance between the digestive effects of the acid and the ability of gastric and duodenal mucosa to resist.

Gastro(duodenal) barrier integrity

Prostaglandins important in protecting mucosa: They stimulate HCO_3 and mucous production, increase mucosal blood flow and modify the local inflammatory reaction to acid.

Acids penetrating the mucosa damage the mast cells releasing histamine and other inflammatory mediators. These mediators cause a local vasodilatation and the tissue reheals when the injury is not too severe. With more severe damage other inflammatory mediators are produced reducing blood flow and resulting in further tissue damage.

Breakdown of the superficial mucosa not involving the submucosa is called an Erosion. Damage to the muscularis mucosa and deeper layers is called ulceration. *H. pylori*, a gram -ve bacillus, colonises the antral mucosa and secretes urease which converts urea into NH_3 . The ammonia buffers stomach acid and protects *H. Pylori* from the acidity. The increase in gastric acid secretion that occurs often in patients with DU's may result from *H. Pylori* antral inflammation inhibiting somatostatin release by the antral D cells and this would result in loss of the inhibition of gastrin. Inhibition of acid secretion heals but doesn't cure *H. Pylori* peptic ulcers. Antibiotics are needed to eradicate the *H. Pylori*.

Zollinger-Ellison syndrome

- Gastrin-secreting tumor (gastrinoma) of pancreas or duodenum.
- Acid hypersecretion causes recurrent ulcers in duodenum and jejunum.
- Presents with abdominal pain (peptic ulcer disease, distal ulcers), diarrhea (malabsorption).
- Positive secretin stimulation test: gastrin levels remain elevated after administration of secretin, which normally inhibits gastrin release.
- May be associated with MEN type1.

MEN type1 and Gastrinoma

Multiple Endocrine Neoplasia (MEN) syndromes are rare inherited disorders characterized by the occurrence of tumors in multiple endocrine glands. They are caused by genetic mutations and are autosomal dominant disorders. There are many types and MEN type1 is one among them.

MEN Type 1:

- **Involves tumors in following 3 endocrine glands (“all begin with P”):**
 - **Parathyroid glands** (Primary hyperparathyroidism):
 - Patient presents with hypercalcemia and kidney stones.
 - **Pancreatic islet cells:**
 - Tumors like gastrinomas (causing Zollinger-Ellison syndrome), insulin-secreting tumors, or glucagon-secreting tumors.
 - **Pituitary gland:**
 - Prolactin-secreting or growth hormone-secreting tumors are the most common pituitary tumors.

Thank you

**For question and clarifications
please email**

Dr Janine Paul

jpaul@sgu.edu or ask-phy@sgu.edu